

Chapter 28 Back Matter Check

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28 Supersymmetric Versions of the Standard Model

Problems

1. Suppose that the interactions (28.1.4) and (28.1.5) were actually present in the Lagrangian of a supersymmetric version of the standard model. Make a rough estimate of how heavy the squarks and sleptons would have to be to avoid a conflict with experimental bounds on the proton lifetime.
2. Suppose that the typical mass m of the gauginos, higgsinos, squarks and sleptons is very much larger than m_Z . Give the renormalization group equations for the running gauge couplings at energies both above and below m . Use the results, together with the unification assumption employed in Section 28.2, to give formulas for $\sin^2 \theta$ and the unification scale M in terms of m , m_Z , $e(m_Z)$, $g_s(m_Z)$, and n_s . How large could m be without violating experimental bounds on $\sin^2 \theta$ and M ?
3. Give formulas for the couplings of the quarks and leptons to the lightest CP-even neutral scalar particle in the minimum supersymmetric standard model in terms of the parameters m_A , m_Z , β , G_F , and the quark and lepton masses.
4. Use holomorphy arguments to derive the one-loop formula for the gluino mass in a theory of gauge-mediated supersymmetry breaking, in which the messenger superfields Φ_n and $\bar{\Phi}_n$ get their masses from a term $\sum_n \lambda_n S_n (\bar{\Phi}_n \Phi_n)$ in the superpotential, in terms of the expectation values $\mathcal{S}_n$ and $\mathcal{F}_n$ of the ϕ - and F -components of the singlet superfields S_n , in the limit where $|\mathcal{F}_n| \ll |\lambda_n| |\mathcal{S}_n|^2$.
5. Taking account of the possibility of a small flavor dependence of squark masses, estimate the gluino exchange contribution to baryon and lepton non-conserving four-fermion interactions among the quarks and leptons. Set an upper bound on these contributions, using the bound on squark mass splittings from the rate of $K^0 \rightarrow \bar{K}^0$ conversion.

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